

Curriculum-vitae

Manoj Kumar Yadav, PhD

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Educational qualifications:

Degree	Year	Board/ University	Division
PhD (Genetics)	2023	Maharshi Dayanand University, Rohtak, India	NA
MSc (Genetics)	2013	Maharshi Dayanand University, Rohtak, India	First
BSc (Zoology)	2011	University of Delhi, New Delhi, India	Second

Research interest:

General: Microbial technology

Specific: Purification, characterization and mode of action of bacteriocins against indicator strains. Antimicrobial activity of bacteriocins using indicator-dependent and indicator-independent (vesicles-based colorimetric assay) methods. Applications of bacteriocins in food-preservation and therapeutics.

Title of PhD thesis: Method optimization for high-throughput screening of bacteriocin-producing lactic acid bacteria. <http://hdl.handle.net/10603/477741>

PhD Supervisor: Dr. Santosh Kumar Tiwari, Professor, Department of Genetics, Maharshi Dayanand University, Rohtak, Haryana, India.

Summary of work done:

My PhD thesis focused on purifying and characterizing bacteriocins from lactic acid bacteria isolated from fermented foods and soil. Two bacteriocins-plantaricin LD1 (*Lactobacillus plantarum* LD1) and weissellicin LM85 (*Weissella confusa* LD85) - showed strong antimicrobial

and antibiofilm activity against multiple pathogens. Their membrane-acting mode was confirmed by K⁺ efflux, loss of membrane potential, disrupted pH gradient, fluorescence staining, and microscopy. Plantaricin LD1 also effectively eliminated *S. aureus* in milk and *S. Typhimurium* in fruit juice. These strains represent promising candidates for food safety and therapeutic applications.

Research Experience:

1. **Post-Doctoral Researcher** from **1 April 2024 to till date** in Department of Animal Biotechnology, Dankook University, Cheonan-si, Chungcheongnam-do, Republic of Korea.
2. **PhD Student** from 30 December 2019 to 17 February 2023 (**3 years 1 months 17 days**) in Department of Genetics, Maharshi Dayanand University, Rohtak-124001, Haryana, India.
3. **Research Assistant** (ICMR project, New Delhi, India) from 7 December 2017 to 30 December 2019 (**2 years and 24 days**) in Microbial Genetics Lab, Department of Genetics, Maharshi Dayanand University, Rohtak-124001, Haryana, India.
4. **Senior Research Fellow** (DBT project, New Delhi, India) from 21 December 2015 to 12 November 2016 (**10 months and 23 days**) in Microbial Genetics Lab, Department of Genetics, Maharshi Dayanand University, Rohtak-124001, Haryana, India.
5. **Junior Research Fellow** (DBT, New Delhi, India) from 20 December 2013 to 20 December 2015 (**2 years**) in Microbial Genetics Lab, Department of Genetics, Maharshi Dayanand University, Rohtak-124001, Haryana, India.

Publications

1. **Yadav MK**, Mendoza RM, Song JH, Vasquez R, Lee JS, Park YS, Paik HD, Kim IH, Kang DK (2025). A fucose-containing hetero-exopolysaccharide from *Levilactobacillus brevis* B151 exhibits potent antimicrobial, antibiofilm, and antioxidant activities. Probiotics and Antimicrobial Proteins, 1-16. <https://doi.org/10.1007/s12602-025-10835-5>. ISSN:1867-1306. (**IF 4.4**).
2. **Yadav MK** and Tiwari SK (2025). Mechanism of action and antimicrobial potential of weissellicin LM85 from *Weissella confusa*. Nutraceuticals, 5(4):33. <https://doi.org/10.3390/nutraceuticals5040033>. E-ISSN:1661-3821.
3. **Yadav MK**, Baldia A and Tiwari SK (2025). Plantaricin LD1 inhibits *Salmonella enterica* subsp. *enterica* serovar Typhimurium ATCC13311 in fruit juice. Biology Bulletin, 52(4):164. <https://doi.org/10.1134/S106235902560117X>. ISSN:1062-3590. (**IF 0.5**).

4. **Yadav MK**, Song JH, Vasquez R, Lee JS, Kim IH, Kang DK (2024). Methods for detection, extraction, purification, and characterization of exopolysaccharides of lactic acid bacteria - A systematic review. *Foods*, 13(22):3687. <https://doi.org/10.3390/foods13223687>. ISSN:2304-8158. **(IF 5.2)**.
5. **Yadav MK** and Tiwari SK (2024). Weissellicin LM85 purified from *Weissella confusa* LM85 effluxes potassium ions and depletes proton motive force in *Escherichia coli* ATCC25922. *International Journal of Peptide Research and Therapeutics*, 30(4):44. <https://doi.org/10.1007/s10989-024-10622-w>. ISSN:1573-3904. **(IF 2.0)**.
6. **Yadav MK** and Tiwari SK (2024). Mechanism of cell killing activity of plantaricin LD1 against *Escherichia coli* ATCC25922. *Applied Biochemistry and Biotechnology*, 196(11):7570-7587. <https://doi.org/10.1007/s12010-024-04927-1>. ISSN:0273-2289. **(IF 3.1)**.
7. Sheoran P*, **Yadav MK***, Kumari I and Tiwari SK (2024). Enterocin LD3 from *Enterococcus hirae* LD3 inhibits the growth of *Salmonella enterica* subsp. *enterica* serovar Typhimurium ATCC13311 in fruit juice. *Probiotics and Antimicrobial Proteins*, 16(4):1205-1213. <https://doi.org/10.1007/s12602-023-10108-z>. ISSN:1867-1306. **(IF 4.4)** (*Equal authorship).
8. **Yadav MK**, Baldia A and Tiwari SK (2024). Plantaricin LD1 inhibits the growth and biofilm formation of *Staphylococcus aureus* in milk. *Journal of Exploratory Research in Pharmacology*, 9(1):1-7. <https://doi.org/10.14218/JERP.2023.00011>. ISSN:2572-5505.
9. **Yadav MK** and Tiwari SK (2023). Methods for determination of antimicrobial activity of bacteriocins of lactic acid bacteria. *Microbiology*, 92(6):745-765. <https://doi.org/10.1134/S0026261723600520>. ISSN:0026-2617. **(IF 1.3)**.
10. **Yadav MK**, Yadav P, Dhiman M, Tewari S and Tiwari SK (2022). Plantaricin LD1 purified from *Lactobacillus plantarum* LD1 inhibits biofilm formation of *Enterococcus faecalis* ATCC29212 in tooth model. *Letters in Applied Microbiology*, 75(3):623-631. <https://doi.org/10.1111/lam.13668>. ISSN:0266-8254. **(IF 2.0)**.
11. **Yadav MK**, Kumari I, Singh B, Sharma KK and Tiwari SK (2022). Probiotics, prebiotics and synbiotics: Safe options for next generation therapeutics. *Applied Microbiology and Biotechnology*, 106(2):505-521. <https://doi.org/10.1007/s00253-021-11646-8>. ISSN:0175-7598. **(IF 3.9)**.
12. **Yadav MK** and Tiwari SK (2021). Polydiacetylene vesicles acting as colorimetric sensor for the detection of plantaricin LD1. *Analytical Biochemistry*, 631:114368. <https://doi.org/10.1016/j.ab.2021.114368>. ISSN:0003-2697. **(IF 2.6)**.
13. **Yadav MK**, Singh B and Tiwari SK (2019). Comparative analysis of inhibition-based and indicator-independent colorimetric assay for screening of bacteriocin-producing lactic acid bacteria. *Probiotics and Antimicrobial Proteins*, 11(2):687-695. <https://doi.org/10.1007/s12602-018-9445-4>. ISSN:1867-1306. **(IF 4.4)**.
14. **Yadav MK**, Kumar V, Singh B and Tiwari SK (2017). Phospholipid/polydiacetylene vesicle-based colorimetric assay for high-throughput screening of bacteriocins and halocins.

Papers presented in conferences:

1. **Yadav MK** and Tiwari SK (2022). Polydiacetylene vesicles-based colorimetric assay for the detection of bacteriocins produced by probiotic lactic acid bacteria (**Oral presentation**). International conference on “Advances and Challenges in Life Science Research” organized by Department of Biochemistry, School of Life Sciences, Vels Institute of Science, Technology & Advanced Studies, Pallavram, Chennai, India on 12th & 13th April, 2022.
2. **Yadav MK**, Yadav P, Dhiman M, Tewari S and Tiwari SK (2020). Plantaricin LD1 as potential alternative to antibiotics in oral health (**Poster presentation**). 5th Biennial conference of PAi and International symposium on probiotics and immunity: way forward to microbial therapy. Organized by probiotic association of India in association with ICAR-national dairy research institute Karnal-132001, India on 19-20th November 2020.
3. **Yadav MK** and Tiwari SK (2019). An indicator-independent colorimetric assay for the detection of membrane-acting bacteriocins (**Oral presentation**). International Symposium on Microbes for Sustainable Development of Energy, Environment, Agriculture and Health. Organized by Central University Haryana, Mahendergarh-123031, India on 15-18th November 2019.
4. **Yadav MK** and Tiwari SK (2019). Probiotic bacteriocin as alternative to clinical antibiotics (**Poster Presentation**). National conference on “Global climate change: Concerns and solutions”. Organized by Environment Sustainability Management Cell, MDU, Rohtak-124001, Haryana, India on 15th -16th February 2019.
5. **Yadav MK** and Tiwari SK (2017). Comparative study of antimicrobial activity of soil isolates of lactic acid bacteria (**Poster Presentation**). International Conference on Microbes health and wealth. Organized by Department of Microbiology and AMI Rohtak-Unit, MDU, Rohtak-124001, Haryana, India on 14th November 2017.
6. **Yadav MK** and Tiwari SK (2017). High throughput colorimetric assay for the screening of bacteriocin-producing lactic acid bacteria (**Poster Presentation**). International Symposium on Microbes for Sustainable Development: Scope and Applications (MSDSA-2017). Organized by Department of Environmental Microbiology, Babasaheb Bhimrao Ambedkar University, Lucknow, Uttar Pradesh-226025, India on 16-19th November 2017.
7. **Yadav MK** and Tiwari SK (2017). Antimicrobial activity of bacteriocins of lactic acid bacteria isolated from fermented foods (**Poster Presentation**). National conference on “food processing for value addition: trends and innovations”. Organized by Department of Food Technology, MDU, Rohtak-124001, Haryana, India on 27th November 2017.
8. **Yadav MK** and Tiwari SK (2016). Vesicle-based assay for the screening of lactic acid bacteria with potential antimicrobial activity (**Poster Presentation**). National Conference on Genetic Diversity and Therapeutic Potential of Natural Products. Organized by Department of Genetics, MDU, Rohtak-124001, Haryana, India on 17th September 2016.

9. **Yadav MK** and Tiwari SK (2015). Optimization of methods for high throughput screening of bacteriocin producing strains of lactic acid bacteria (**Poster Presentation**). 56th Annual Conference of Association of Microbiologist of India (AMI 2015) and International Symposium on Emerging discoveries in Microbiology. Organized by School of Life Science, JNU, New Delhi-110067, India on 7-10th December 2015.
10. **Yadav MK** and Tiwari SK (2014). High throughput screening of bacteriocin producing lactic acid bacteria (**Poster Presentation**). National Conference on Therapeutic Potential of Natural Products: Current Innovations and Future Trends. Organized by Department of Genetics, MDU, Rohtak-124001, Haryana, India on 19th March 2015.

Memberships:

1. **Life member** of “The Association of Microbiologists of India (AMI)” India (Membership No. 4855-2019).

Award and honors:

1. Received 2nd prize for best poster presentation in National conference on “Global climate change: Concerns and solutions”. Organized by Environment Sustainability Management Cell, MDU, Rohtak-124001, Haryana, India on 15-16th February 2019.

Personal Information:

Name	: Manoj Kumar Yadav
Date of Birth	: 20 July 1992
Gender	: Male
Marital status	: Unmarried
Father's name	: Mr. Ram Sundar Yadav
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Languages Known	: Hindi, English

REFERENCES:

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I hereby declare that the above written particulars are true and correct to the best of my knowledge and belief. The documents in the support of information will be provided when required.

Date:

Place: Cheonan, Republic of Korea



(Manoj Kumar Yadav)